

Recommendation System Models

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Information Table:

<u>Reference (Year)</u>	<u>Description</u>	<u>Advantages</u>	<u>Disadvantages</u>	<u>References</u>
Hidasi et al., <i>GRU4Rec</i> (2016)	Introduced recurrent neural networks (GRUs) for session-based recommendation , trained with pairwise ranking losses. It had modeled sequential dependencies in user interactions, enabling effective next-item prediction in clickstreams and e-commerce.	- Successfully modeled session dynamics and short-term sequential behavior. - Outperformed traditional CF for session-based tasks. - Demonstrated value of deep learning for RS.	- High computational cost compared with MF. - Performed poorly for long sequences or irregular sessions. - Did not handle cross-session personalization.	https://arxiv.org/pdf/1511.06939
He et al., <i>Neural Collaborative Filtering</i> (2017)	Proposed NCF, replacing the dot product in MF with multilayer perceptrons to learn non-linear user-item interaction functions. It had introduced GMF + MLP fusion (NeuMF) and had shown better performance than linear MF on implicit feedback datasets.	- Captured complex non-linear relations. - Unified neural and latent factor methods. - Demonstrated the extensibility of deep models to RS.	- Training was more resource-intensive. - Improvements depended on large, rich datasets. - Risked overfitting under sparsity.	https://arxiv.org/pdf/1708.05031
Rendle, <i>Factorization Machines</i> (2010)	Proposed Factorization Machines (FM), which generalized matrix factorization	- A flexible framework that had worked with heterogeneous feature types.	- Relied heavily on feature engineering. - Did not capture non-linear	https://www.ismll.uni-hildesheim.de/public/pdfs/Rendle2010FM.pdf

	to arbitrary feature spaces. The model represented any sparse input as a vector and factorized pairwise feature interactions. FMs had been widely applied in recommendation because they allowed side information (demographics, metadata, context) to be used alongside implicit feedback.	- Scalable for high-dimensional sparse data (web-scale). - Had unified linear models and MF into a single general form.	higher-order feature interactions without extensions. - Risk of overfitting if features were not regularized.	
Kang & McAuley, <i>SASRec</i> (2018)	First to apply Transformer self-attention to sequential recommendation. Modeled both short- and long-range dependencies in user histories. SASRec had masked attention to predict the next item, showing superior scalability and accuracy compared to RNNs.	- Parallelized training (faster than RNNs). - Captured longer dependencies effectively. - Easily extensible to other sequence tasks.	- Required large datasets to train well. - Prone to overfitting on sparse histories. - Depended on positional encoding design.	https://arxiv.org/pdf/1808.09781
Sun et al., <i>BERT4Rec</i> (2019)	Adapted bidirectional Transformer (BERT) for sequential recommendation using a masked item prediction objective . It had allowed the model to learn context from both left and right of the masked position.	- Rich contextual representations improved sequential prediction. - Outperformed unidirectional models in many domains. - Introduced transfer of NLP pretraining ideas to RS.	- Training and inference were computationally heavy. - Masking design required care to avoid information leakage. - Large model size was impractical for smaller platforms.	https://arxiv.org/pdf/1904.06690
Wang et al., <i>Neural Graph Collaborative Filtering (NGCF)</i> (2019)	Proposed GCN-based collaborative filtering by propagating	- Explicitly modeled graph structure of interactions. - Improved	- Graph propagation was costly in memory and time. - Sensitive to depth	https://arxiv.org/pdf/1905.08108

	embeddings on the user-item bipartite graph, thereby capturing higher-order connectivity and refining embeddings layer by layer.	accuracy by leveraging multi-hop connectivity. - Inspired many follow-up GNN models.	and hyperparameters. - Still limited in handling cold-start.	
He et al., <i>LightGCN</i> (2020)	Simplified NGCF by removing transformations and nonlinearities , keeping only weighted graph neighborhood aggregation. Showed that simplicity improved performance while reducing complexity.	- Computationally efficient and scalable. - Outperformed more complex GCNs in many benchmarks. - Easy to implement and train.	- Still dependent on historical interactions (cold-start unresolved). - Lacked the ability to incorporate side features directly. - Performance is sensitive to the layer combination strategy.	https://arxiv.org/pdf/2002.02126
Wu et al., <i>Self-supervised Graph Learning (SGL)</i> (2021)	Introduced contrastive self-supervised learning for graph-based recommenders. Created augmented graph views (edge/node dropout, random walks) and added a contrastive objective alongside supervised training.	- Boosted performance under data sparsity. - Improved robustness to noisy interactions. - Addressed popularity bias to some extent.	- Augmentation strategies required careful design. - Computational overhead increased. - Could remove useful information if augmentations were too aggressive.	https://arxiv.org/pdf/2010.10783
Xie et al., <i>CL4SRec</i> (2020)	Brought contrastive learning into sequential RS. Created different augmented sequence views (masking, cropping, reordering) and optimized contrastive objectives together with next-item prediction.	- Provided extra supervision signals, alleviating sparsity. - Improved generalization and robustness. - Worked across multiple sequence encoder backbones.	- Augmentations were task-specific, not always generalizable. - Added hyperparameters for contrastive losses. - Computationally more demanding.	https://arxiv.org/pdf/2010.14395
Yu et al., <i>SimGCL</i> (2022)	Proposed a lightweight contrastive learning	- Extremely simple and efficient.	- May not capture structural graph properties.	https://arxiv.org/pdf/2209.02544

	method for GCN-based recommendation. Instead of structural augmentations, applied random noise to embeddings to generate views.	- Achieved comparable performance to heavy augmentation methods. - Easier to reproduce across datasets.	- Sensitivity to noise intensity hyperparameter. - Lacked theoretical guarantees on robustness.	
Zhao et al., <i>RecBole Framework</i> (2020, 2022)	Open-source library supporting dozens of recommendation models with unified interfaces, datasets, and evaluation metrics. Improved reproducibility and benchmarking in RS research.	- Standardized experimentation. - Reduced implementation burden for new researchers. - Facilitated fair comparison of models.	- Limited flexibility for highly customized models. - Depended on preprocessing decisions. - Large-scale deployment outside research required adaptation.	https://arxiv.org/pdf/2011.01731 https://arxiv.org/pdf/2206.07351
Ying et al., <i>PinSAGE: Graph Convolutional Neural Networks for Web-Scale Recommender Systems</i> (2018)	Proposed PinSAGE, a scalable graph convolutional network for large-scale recommender systems (Pinterest). Combines random walks with graph convolutions to generate node embeddings efficiently.	- High scalability (3B nodes, 18B edges). - Captures graph structure + node features. - Strong performance in offline metrics & A/B tests.	- High computational cost for very large graphs. - May require substantial engineering to deploy at scale.	https://cs.stanford.edu/people/jure/pubs/pinsage-kdd18.pdf
Anand & Maurya, <i>A Survey on Recommender Systems Using Graph Neural Networks</i> (2024)	Surveyed the integration of knowledge graphs in recommendation. Categorized methods, discussed cold-start & sparsity issues, and emphasized explainability and semantic reasoning.	- Enhanced recommendation accuracy. - Improved explainability. - Helps cold-start & sparse data problems.	- Complexity in constructing/maintaining knowledge graphs. - Knowledge graphs may be incomplete or outdated.	https://dl.acm.org/doi/pdf/10.1145/3694784
Zhang et al., <i>Recommending on Graphs: A Comprehensive Review from a Data Perspective</i> (2023)	Reviewed graph-based recommendation systems focusing on types of graphs, data handling, and	- Comprehensive categorization of graph types. - Highlights challenges and	- Mostly survey; does not propose new models. - Practical implementation	https://arxiv.org/pdf/2212.12230

	challenges like scalability, fairness, and explainability.	potential research directions. - Data-centric perspective.	guidance limited.	
Syed et al., <i>Context-Aware Explainable Recommendation Based on Domain Knowledge Graph</i> (2022)	Developed a context-aware recommender leveraging domain knowledge graphs for explainable recommendations. Uses logical reasoning on incomplete triples to generate explanations.	- Provides explainable recommendations. - Uses domain knowledge for richer context. - Handles complex user queries	- Limited to domain-specific graphs. - May not scale to very large datasets. - Requires a high-quality domain knowledge graph.	https://www.mdpi.com/2504-2289/6/1/1 1